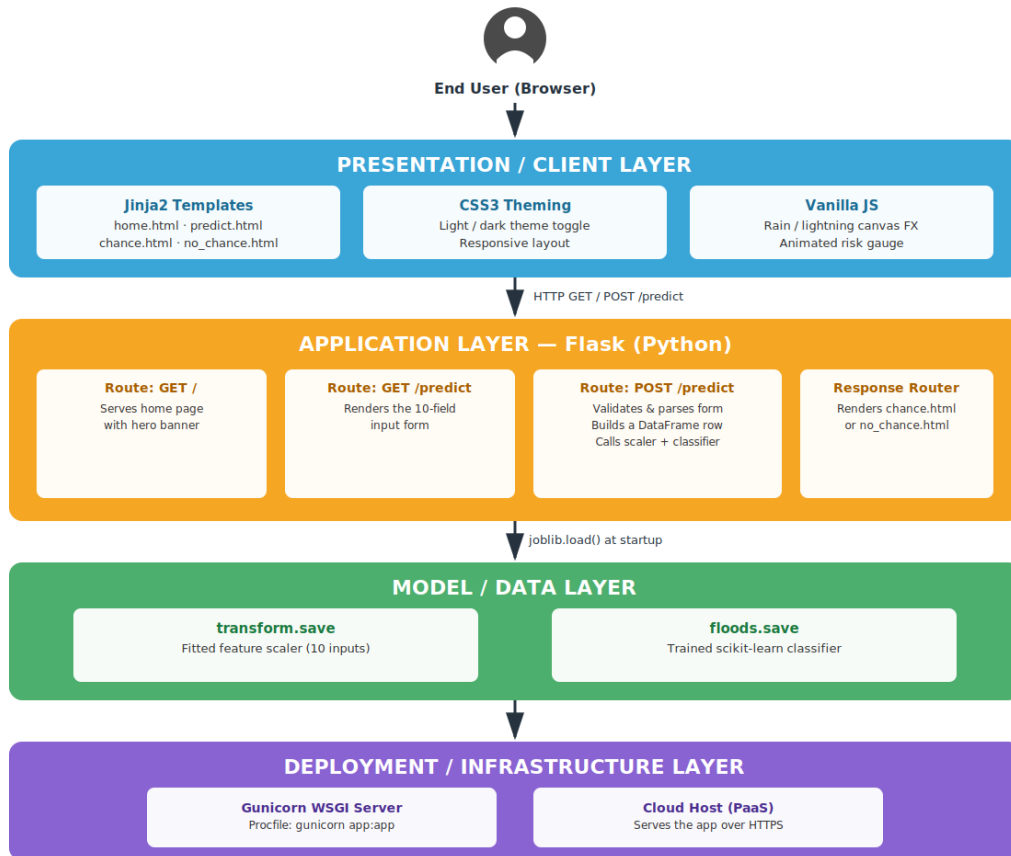


Solution Architecture

Date	04 July 2026
Team ID	
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

SOLUTION ARCHITECTURE — Rising Waters Flood Prediction System



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the form and result pages the user interacts with; drives the theme toggle and animated risk gauge.	HTML5, CSS3, Vanilla JS
Application Layer	Flask routes that handle requests, validate form data, and orchestrate scaling + prediction.	Python, Flask
Model / Data Layer	Holds the trained classifier and its fitted feature scaler; performs the actual inference.	Scikit-learn, Joblib
Production Server	Serves the Flask app under real traffic instead of the dev server.	Gunicorn (Procfile: gunicorn app:app)